

EE4437 Homework 2 (Part I)

Due: 22 Mar 2026, 11:59 pm

Please submit your answers with justification as well as program codes (e.g. Python) if any.

- Q1. A quantum well laser is achieved by stacking multiple GaAs, $\text{Al}_{0.2}\text{Ga}_{0.8}\text{As}$ layers. The width of the well is found to be 10 nm. Find the 3 lowest energy values of lasing photons using an infinite well approximation. Consider both light holes and heavy holes. Use the material parameters in page 56 of Chapter 1. Make sure that the eigenenergies are confined by the $\text{Al}_{0.2}\text{Ga}_{0.8}\text{As}$ layer using the band offsets in page 54 of Chapter 1. $m_0 = 9.11 \times 10^{-31}$ kg, and $\hbar = 1.055 \times 10^{-34}$ J.s. (20 pt)
- Q2. (a) A III-V semiconductor has bandgap of 1.3 eV. Calculate the Fermi levels if we have $5 \times 10^{19} \text{ cm}^{-3}$ electrons injected. Use the electron effective mass of $0.055m_0$. $m_0 = 9.11 \times 10^{-31}$ kg, and $\hbar = 1.055 \times 10^{-34}$ J.s. (5 pt)
- (b) Assume the valence band consists only of the heavy hole band with the heavy hole effective mass of $0.5m_0$ and $5 \times 10^{19} \text{ cm}^{-3}$ of holes are injected. Calculate the valence band Fermi level. (5 pt)
- (c) Plot the net absorption coefficient versus the energy ($\hbar\omega - E_g$) in eV unit at 10 K and 300 K. The y-axis should be arb. units. The Boltzmann constant (k_B) is $1.38 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$. (12 pt)
- (d) Calculate the width of the gain region (in energy). (6 pt)
- (e) Repeat (c) with 10^{20} cm^{-3} of holes. Other conditions are the same. Discuss the change in the net absorption spectrum compared to the case in (c) and explain the origin of those differences. (Hint: plot the occupation factors of electron and hole separately at different temperatures). (12 pt)
- Q3. $2 \times 10^{18} \text{ cm}^{-3}$ holes and electrons are injected into the GaAs by optical absorption from a laser beam. Plot the spontaneous emission spectrum as a function of the energy ($\hbar\omega - E_g$) in eV unit at room temperature. The y-axis should be arb. units. There are no mirrors in the structure so the structure cannot lase. Take the GaAs bandgap energy, $E_g = 1.519 \text{ eV}$, with an electron mass $m_e = 0.0665m_0$, and a heavy hole mass of $m_{hh} = 0.45m_0$, where $m_0 = 9.11 \times 10^{-31} \text{ kg}$ is the free electron mass. Note that $[\hbar\omega]^2$ in the formula of the spontaneous transition rate per unit volume per unit energy can be approximated as a constant of E_g^2 . You may neglect the light hole band for simplicity, and neglect any reabsorption of any spontaneous emission by GaAs. You may also neglect quantum confinement effects. $\hbar = 1.055 \times 10^{-34} \text{ J.s}$. (20 pt)